TECHNICAL IMPLEMENTATION REPORT

Project Presentation Dev. Foundation Phase

SHORT DESCRIPTION

Pure Truth Water Factory — Business Process Automation with Salesforce

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1. Executive Summary

Pure Truth Water Factory completely modernized its internal employee loan management through the deployment of a native Salesforce application, built on the Aetotech Loan App Mgr managed package and automated end-to-end with Flow Builder and Approval Processes.

The system's architecture was structured around three main business processes — employee onboarding (Employee Onboarding), loan request management (Loan Application), and repayment processing (Loan Repayment) — complemented by an automated module for closing the loan lifecycle and analytical reporting capabilities. In this report the three processes are documented as Topic 1, Topic 2 and Topic 3.

The entire circuit has been fully validated in a real operational environment through end-to-end testing sessions with genuine users, ensuring data security and the operational effectiveness of the solution.

1.1 Naming Guide and Glossary

The table below shows how each business process maps to the Salesforce components named throughout this report.

Process (Topic)	Performed by	Screen Flow	Entry point	Approval Process
Employee Onboarding (Topic 1)	Finance Officer	Employee Onboarding	Run directly by the Finance Officer	None
Loan Application (Topic 2)	Employee	Loan Application Request	Quick Action "Request a Loan" on the Account	Loan Application Approval
Loan Repayment (Topic 3)	Employee	Submit a Repayment	Quick Action "Submit a Repayment" on the Account	Loan Repayment Receipt: Submit Loan Repayment for Approval

Term	Meaning in this project
Screen Flow	A Flow with screens that a user fills in step by step. Runs in User Context, with the permissions of the person using it.
Autolaunched Subflow	A Flow without screens, called from another Flow to perform one specific task (here, writing a field the user cannot edit).
Record-Triggered Flow	A Flow that runs automatically when a record is created or updated, with no user action.
Approval Process	Salesforce's native feature to route a record to an approver, with approval history, notifications and final actions.
FLS (Field-Level Security)	The setting that decides, per Profile, whether a field is hidden, read-only or editable.
OWD (Organization-Wide Defaults)	The baseline setting that decides who can see each record of an object.
System Context With Sharing	Mode in which a subflow ignores Object Permissions and FLS but still respects record sharing.

2. Business Case and Requirements

2.1 Coverage Matrix — Challenges With Current System

#	Original problem	Applied solution	Status
1	Difficult to calculate/report the total amount loaned	Custom Report Type + "Loan Portfolio Summary"	Covered
2	Only 1 person can edit the Excel sheet at a time	Native Salesforce multi-user access	Covered (inherent)
3	Bank receipts get lost	File Upload attached to the record (Topic 3)	Covered
4	Bank statement printed once a month	Online verification against the bank upon approval	Covered
5	No limit validations or multiple-loan checks	40% Loan Limit + duplicate check (Topic 2)	Covered

2.2 Coverage Matrix — Management Requirements

#	Requirement	Applied solution	Status
1	Calculate/report loan issued, interest earned, balance, outstanding	Loan Application fields + "Total Interest Amount" + reports	Covered
2	Simultaneous multi-user use	Inherent to Salesforce	Covered
3	Store a copy of the bank receipt	File Upload (Topic 3)	Covered
4	Weekly/monthly repayment reports	Weekly Report Subscription	Covered
5	Validate multiple requests and loan limit	Duplicate check + Loan Limit (Topic 2)	Covered

2.3 Coverage Matrix — Additional Requirements

Process	Requirement	Applied solution	Status
Onboarding	Handled by Finance Officer, license restricted by Profile	Screen Flow + Profile-based access restriction	Covered
Loan Application	Handled remotely by any employee	Quick Action on Account + Employee Self-Service	Covered
Loan Repayment	Remote, approved by Finance Officer	Quick Action + Approval Process (Lewis Webb)	Covered
Security	User login to restrict who can apply for loans	Differentiated Profiles + FLS + Flow access management	Covered

3. Solution Architecture

3.1 Roles and Profiles

- **Marisa Roberts** — System Administrator (admin account).
- **Angela James** — Profile Employee Self-Service, Role Employee. Test user with a real license, always tested with a genuine login.

- **Lewis Webb** — Profile Finance Officer, Role Loan Officer Role. Approves both Loan Applications and Loan Repayment Receipts.

Methodological decision: all system testing was carried out using the genuine credentials of Angela James and Lewis Webb, deliberately avoiding the Administrator's "Login As" feature. This decision ensures that the validated behavior — including Field-Level Security (FLS) and Profile restrictions — is exactly what the end user experiences, rather than an approximation from an Administrator's elevated context.

Note — "Finance Manager" Profile

As part of the org's initial setup (Task 1001 of the case setup guide), two Profiles were cloned from Standard User: "Finance Officer" and "Finance Manager," along with the Role hierarchy CFO → Loan Manager Role → Loan Officer Role (Task 1002).

The original instructions only requested the creation of one active test user, with Profile "Finance Officer" and Role "Loan Officer" (Task 1003) — that is Lewis Webb.

The "Finance Manager" Profile is part of the org's setup/Role hierarchy architecture, with no active business user assigned to it and no Approval Process referencing it within this scope — it does not participate as an operational actor in any tested process. It is documented here to make explicit that Task 1001 was fully completed, leaving no unexplained gap.

Evidence — Role Hierarchy and Actual User Assignment

The org's Setup confirms the configured Role hierarchy (Pure Truth Water Factory → CEO → CFO → Loan Manager Role → Loan Officer Role) and the actual assignment of Lewis Webb to the "Loan Officer Role," with his genuine corporate user and email (lewis.web@puretruth.com) — evidence that all Approval Process testing was performed with a real organization user, never simulated.

3.2 Account ↔ User Link

A custom field, Employee_User__c (Lookup from Account to User), was created, allowing any Flow to resolve "who is logged in" to "which employee record they correspond to," via a Get Records filtering by Employee_User__c Equals $User.Id — without relying on the employee self-identifying manually, which would have opened the door to identity impersonation.

3.3 3-Layer Security Model

Layer	What it controls	Does System Context bypass it?
1. Object Permissions	Access to the object in general	Yes, in both variants
2. Field-Level Security (FLS)	Access to a specific field	Yes, in both variants
3. Sharing / OWD (Organization-Wide Defaults)	Access to a record based on its Owner	Only the "Without Sharing" variant

Project philosophy: restrict, don't expand. Each person receives only the permissions they need for their business role.

3.4 Data Model — Overview

The system relies on three custom objects and a set of linking fields on Account, along with numerous formula fields and Roll-Up Summary fields that handle business calculations (balance, interest, limits). The

field-by-field detail, including the exact formulas, is documented in Appendix A — Data Dictionary, at the end of this report, to avoid overloading the main body of the document.

- **Loan_Application__c** — records each loan request and its complete lifecycle (Submitted → Approved/Rejected → Paid Off), along with the calculated amounts (principal, interest, balance).
- **Loan_Repayment_Receipt__c** — records each individual payment (Master-Detail of Loan Application), with its bank receipt and its own approval cycle (Draft → Submitted → Approved/Rejected).
- **Salary_Scale__c** — catalog of salary scales by Designation, used to validate the 40% loan limit.
- **Account** (standard Salesforce object, repurposed as the employee record via custom fields) — linked to the employee's real User (Employee_User__c) and to their Salary Scale (Designation__c), enabling secure and automatic identification of "who is requesting the loan."

3.5 Scope — What's Outside the Automated System

The original (manual, pre-project) process included, as the final step of loan approval, the Finance Officer transferring the actual funds to the employee's bank account. That actual bank transfer is outside the scope of this system: Salesforce does not move real funds without an external payment integration, and none of the case's Management Requirements or Additional Requirements ask for the automation of money movement — they ask to manage, calculate, validate, and report.

The system built covers the complete data and approval cycle (request → validation → approval → repayment recording → verification → loan closure), leaving the actual bank execution as a subsequent manual action, just as in the original process — the difference is that now that manual action is backed by reliable, traceable, and auditable data, instead of an error-prone Excel spreadsheet.

3.6 Visual Identity and Corporate Branding

Requirement / Need: Ensure the platform reflects the institutional identity of Pure Truth Water Factory consistently across the board, without altering the business logic or the specific behavior of the loan application.

Applied solution: Configuration of a global custom theme via Themes and Branding, incorporating the company's official logo optimized to a horizontal ratio of 600x120 pixels for correct display in the Salesforce header.

Why this solution: It standardizes the organization's visual experience end-to-end, ensuring a consistent brand identity for all active users (such as Angela James and Lewis Webb) while operating within the production environment.

Result / Verification: Theme applied and visually validated in the organization's navigation interface, showing the institutional logo in the platform's upper-left panel.

3.7 Reused Security Pattern: System Context Subflow as a Deliberate FLS Bypass

Problem context: The 3-layer security model described in Section 3.3 imposes a real constraint on Screen Flow design: when a Screen Flow runs in User Context (as is appropriate for a flow operated by Employee Self-Service), it is subject to the Field-Level Security of the profile executing it. This means that if a field is configured as Read-Only for that profile, the Flow cannot write to it — regardless of whether the Flow's own business logic is what determines that value.

While building the "Loan Application Request" Flow — the first in the project to present this FLS issue, though not the first Flow built (that was Employee Onboarding, Topic 1) — the Loan_End_Data__c field had an FLS restriction for the Employee Self-Service profile. This root cause is documented in the technical description of the subflow created to resolve it ("Creates the Loan Application record securely using system context to populate the calculated Loan End Data field"). The applied solution was not to relax the profile's FLS to allow direct writing — that would have contradicted the "restrict, don't expand" principle already stated in 3.3 — but rather to extract the entire record creation into an Autolaunched Subflow run in System Context With Sharing, invoked from the main Screen Flow.

This same pattern was recognized as reusable and applied again, without needing to solve the problem from scratch, in later cases within Topic 3:

Location	Applied subflow	FLS-protected field	Object
Topic 2 — Loan Application	Loan Application - Create Secure (V2)	Loan_End_Data__c	Loan_Application__c
Topic 3 — Loan Repayment	Create Loan Repayment Receipt (Draft) (V1)	Status__c → Draft	Loan_Repayment_Receipt__c
Topic 3 — Loan Repayment	Update Loan Repayment Status to Submitted (V1)	Status__c → Submitted	Loan_Repayment_Receipt__c
Topic 3 — Loan Repayment	Confirm Repayment No Receipt (V1)	Status__c → No Receipt	Loan_Repayment_Receipt__c

Why "With Sharing" was preserved instead of "Without Sharing": The deliberate choice of With Sharing (rather than Without Sharing) across the four subflows is consistent with the table in Section 3.3: both variants of System Context bypass Object Permissions and FLS (layers 1 and 2), but only Without Sharing also bypasses Sharing/OWD (layer 3). Using With Sharing means the subflow exclusively resolves the specific FLS issue that caused it, without accidentally expanding the scope of which records the process can see or affect.

Result: This pattern was identified once, during the development of Topic 2, and was consciously and consistently reapplied in Topic 3 when facing the same type of blocker, instead of solving each case with a different solution. This deliberate reuse is evidence that the FLS restriction was not worked around through trial and error, but resolved through an understood and repeatable architectural pattern.

4. Topic 1 — Employee Onboarding

Screen Flow that allows creating the Account for each employee, with duplicate validation by Employment Number, linking to their Salary Scale and their Salesforce User.

4.1. Duplicate employee records due to human error

Link to client requirement: Additional Requirement — Employee Onboarding, handled by Finance Officer.

Symptom: The previous system had no automatic control to prevent the same employee from being loaded twice.

Root cause: Manual data entry with no validation.

Applied solution: Screen Flow with Get Records by Employment Number + Decision that blocks creation if the employee already exists.

Why this solution: Blocking before creation prevents the duplicate from being generated, rather than detecting and cleaning it up afterward.

Result / Verification: Tested with an existing number (blocked) and a new number (Account created). Active.

5. Topic 2 — Loan Application Request

This is where any employee can request a loan from their own Account, with automatic business validations, and automatic submission for Finance Officer approval.

5.1. Employees could accumulate multiple active loans

Link to client requirement: Challenge #5 — "An employee can collect multiple loans."

Symptom: The previous system did not prevent a second request while the first was still in progress.

Root cause: Total absence of control in the manual process.

Applied solution: Get Records + Decision that blocks a new request if one already exists with a Status other than Rejected (later expanded to also exclude Paid Off).

Why this solution: Addressing the issue at the moment of the request prevents the invalid loan from ever being created.

Result / Verification: Tested with a second attempt on a Submitted/Approved loan → blocked.

5.2. Employees could request loans beyond their repayment capacity

Link to client requirement: Challenge #5 — loan limit relative to salary.

Symptom: There was no validation of the loan limit against the employee's salary.

Root cause: Manual process with no automatic calculation or cross-check against the Salary Scale.

Applied solution: Formula Resources within the Flow: estimated installment compared against 40% of gross salary (Gross_payment__c).

Why this solution: The calculation is done before saving the record, because the object's actual formula field does not yet exist at that point.

Result / Verification: Tested with an exceeded amount (blocked) and a valid amount (proceeds).

5.3. The employee could not request the loan remotely and securely

Link to client requirement: Additional Requirement — remote process + "user login page" security.

Symptom: The process depended on manual entry by Finance; additionally, any user with general "Run Flows" permission could have run the Flow without business control.

Root cause: Without an explicit restriction, a Screen Flow is not automatically limited by Profile.

Applied solution: "Request a Loan" Quick Action on Account + explicit Flow restriction enabled only for Employee Self-Service.

Why this solution: Applies the principle of least privilege, meeting the security requirement of the case.

Result / Verification: Tested with Angela James's real login, end to end.

5.4. Submission for approval depended on a manual action

Link to client requirement: Additional Requirement — reduce Finance's manual workload.

Symptom: A remote, automated process would lose its purpose if someone had to manually find a submit button.

Root cause: The native Approval Process requires someone to trigger the submission.

Applied solution: Within the same creation subflow, a "Submit for Approval" step was added, pointing to the newly created record.

Why this solution: Consistency with the rest of the remote, automated design; the object didn't even expose that button in the employee's UI.

Result / Verification: The record automatically arrives in Lewis's queue, with no additional action from Angela.

6. Topic 3 — Loan Repayment Process

Screen Flow that allows the employee to upload their bank receipt and request the recording of a repayment, approved by the Finance Officer. Together, these automations allow the employee to upload their bank receipt and request that the payment be recorded, which Lewis Webb approves or rejects. Its components:

- **Submit a Repayment** (V14, Screen Flow): allows the employee to record a repayment on their approved loan, with a mandatory bank receipt, automatically submitting it for review.
- **Create Loan Repayment Receipt (Draft)** (V1, Autolaunched Subflow): creates the receipt record with Status = Draft, resolving the same FLS restriction on Status__c documented in Section 3.7.
- **Update Loan Repayment Status to Submitted** (V1, Autolaunched Subflow): updates the receipt's Status to Submitted once the attached receipt is confirmed.
- **Confirm Repayment No Receipt** (V1, Autolaunched Subflow): updates the receipt's Status to No Receipt once it's confirmed that no receipt was attached.
- **Loan Repayment Receipt - Block Manual Creation** (V3, Record-Triggered Flow): prevents unauthorized manual creation of receipts, throwing a custom error when they are created outside the "Submit a Repayment" flow.

6.1. Payment verification depended on the monthly printed bank statement

Link to client requirement: Challenge #4 — delay caused by the bank statement printed once a month.

Symptom: Finance could not confirm an actual payment until the monthly printed statement.

Root cause: A legacy technological limitation, not a real business requirement.

Applied solution: The employee uploads the receipt; the Loan Officer verifies it online against the bank when resolving the Approval Request, without depending on the printed statement.

Why this solution: Online banking access is available to any organization using Salesforce; it removes the dependency on the monthly cycle with no additional automation needed.

Result / Verification: Solves two challenges at once in a single solution: verification delay (#4) and lost receipts (#3).

6.2. Repayment requests could move forward without an attached bank receipt

Link to client requirement: Management Requirement #3 — Store a copy of the bank receipt for each repayment.

Symptom: The Flow allowed reaching the submit-for-approval stage even if no file had been uploaded.

Root cause: The contentDocIds field of the File Upload component initializes as an empty collection when no file is attached, not as a null value — a behavior not explicitly documented in Salesforce's standard help for this component.

Applied solution: The validation operator was replaced from "Is Null" to "Is Empty" in the corresponding Decision.

Result / Verification: Confirmed that it now correctly blocks submission and displays the missing-receipt message.

6.3. Operational visibility into repayments with no attached receipt

Link to client requirement: Management Requirement #3 — Store a copy of the bank receipt for each repayment; and reducing the Finance Officer's manual workload.

Applied solution: The value "No Receipt" was added to the Loan Repayment Receipt's Status picklist, automatically updated by a dedicated subflow ("Confirm Repayment No Receipt") when an employee attempts to record a payment without attaching the mandatory bank receipt.

Why this solution: Without this distinction, an upload attempt with no receipt would be indistinguishable from a legitimate repayment under review, forcing the Loan Officer to spend time evaluating an incomplete request. By identifying it with its own status, these cases are filtered out of the approval circuit at the source, and are directly visible in reports and the Dashboard for operational tracking.

Result / Verification: Confirmed with real evidence — an upload attempt with no receipt from Angela James was correctly reflected as "No Receipt" in the record, in the "Loan Repayment Detail Report," and in the Dashboard.

6.4 Deliberate design in the Repayment Approval Process

Record Editability restricted to Administrator

Link to client requirement: Consistency of the approval architecture between Topic 2 and Topic 3.

Symptom: Unlike the Loan Application approval process (Topic 2), where the Record Editability field required configuring "Administrators OR the currently assigned approver," in the "Submit Loan Repayment for Approval" process this parameter was set to "Administrator ONLY."

Root cause: This is not an architectural oversight. In Topic 2, Lewis needed to view and modify additional fields exposed directly on the approval screen. In contrast, Topic 3's review circuit is limited exclusively to auditing the attached receipt against the loaded amount and issuing an Approve/Reject decision, with no need to manually edit any field on the record while it remains locked.

Applied solution: The "Administrator ONLY" option for the Record Editability parameter was intentionally kept. The action of approving or rejecting is inherent to the role of the approver assigned in the flow step,

and does not depend on this global edit permission (which only governs the manual alteration of additional fields).

Why this solution: Applies the principle of least privilege ("restrict, don't expand"), avoiding granting unnecessary permissions for a function that does not require them. It allows two independent layers to coexist harmoniously:

- The Finance Officer profile's base permissions, which grant Lewis general read and edit access to receipts for day-to-day operation.
- The approval process's temporary lock (Record Editability + Record Lock), which suspends free manual editing exclusively while the record is under review, subordinating any change to the circuit's decision buttons.

Result / Verification: Validated with real evidence in the organization: Lewis Webb successfully approved record LRR-000008 (adding his analytical comment) using the Approve button in the approval history, empirically confirming that the "Administrator ONLY" configuration does not restrict the assigned approver's operational capacity for their specific task.

7. Loan Lifecycle Closure

7.1. No control existed to distinguish between an active loan and one already paid off

Link to client requirement: Unforeseen consequence of Challenge #5's own control.

Applied solution: "Paid Off" value added to the picklist + Record-Triggered Flow "Auto Close on Full Repayment" (Loan_Balance__c ≤ 0 AND Status ≠ Paid Off).

Why this solution: A Flow cannot write a picklist value that does not formally exist.

Result / Verification: Flow activated; corresponding adjustment in Topic 2 to enable legitimate new requests.

8. Reporting

The system implements advanced analytical capabilities to give management visibility into the portfolio's financial status and to enable operational control of repayments. To ensure governance and data security, all analytical assets are hosted in a dedicated custom folder (Loan Management Finance Reports) rather than public directories, with access explicitly regulated through role-based sharing rules restricted to authorized profiles (such as the Loan Officer Role).

To meet these business intelligence needs, two main analytical tools were structured:

8.1. Report: Loan Portfolio Summary

What it does: Consolidates macro-level loan information, showing each loan once at the header level. It is structured in Summary format, grouped by status (Status), and includes automatic subtotals for principal issued, interest earned, amount paid, and outstanding balance.

Link to requirement: Challenge #1 — historical difficulty calculating and auditing the total loaned amount in the previous Excel spreadsheet.

Applied solution: Creation of a combined Custom Report Type, a filter excluding Status = Rejected (a rejected loan was never disbursed and should not count as issued or outstanding capital), and assignment of the formula field “Total Interest Amount” (Total_Loan_Repayment__c – Loan_Amount__c), ensuring absolute consistency with the amounts validated by the business engine.

Why this solution: A custom report type is essential to cross data from both objects and prevent the parent’s metrics from being artificially multiplied by each child row.

Result / Verification: Report stored in Loan Management Finance Reports, with exact financial subtotals and no duplication.

8.2. Report: Loan Repayment Detail Report

What it does: Details each payment receipt (Loan Repayment Receipt) at the transactional level, with its payment date, amount, channel, and validation status, facilitating daily or operational auditing.

Link to requirement: Management Requirement #4 — Verification and control of recorded payments.

Applied solution: Strict analytical separation from the portfolio report to resolve the structural ambiguity caused by both objects having fields named Status with different purposes.

Why this solution: Separating them serves the different levels of analysis.

Result / Verification: Clean, ordered transactional listing aligned with cash and bank controls.

8.3. Cleanup of empty groups via advanced relational filters

Link to requirement: Precision and cleanliness in the presentation of analytical data.

Symptom / Challenge: Initial reports showed residual empty categories (indicated with dashes -) for requests that did not yet have associated receipts.

Applied solution: Implementation of a cross filter (Cross Filter — “Loan Applications — with — Loan Repayment Receipts”), the technical equivalent of a relational EXISTS query in databases.

Why this solution: The existence or non-existence of a related record must be evaluated at the relationship structure level, not through conditional filters on blank fields.

Result / Verification: Elimination of fictitious groupings, ensuring a clean and consistent total count in the associated reports and dashboards.

8.4. Automation and periodic distribution via subscriptions

Link to requirement: Management Requirement #4 — Periodic (weekly/monthly) generation of repayment reports by the system.

Symptom / Challenge: The platform allowed on-demand manual queries, but required an autonomous mechanism to deliver information to management or operational roles.

Applied solution: Configuration of a weekly Report Subscription on the key reports, scheduled independently under Lewis Webb’s management profile and credentials.

Why this solution: Enabling native subscriptions removes the need for manual intervention to extract information, facilitating automated periodic follow-up directly in authorized users’ inboxes.

Result / Verification: Subscription confirmed as active in the system with its corresponding automatic execution schedule.

9. Dashboard

Dashboard: Loan Management Overview

As a visual complement to the analytical reports, a centralized executive dashboard was configured within the secure Loan Management Finance Reports folder — restricted through role-based sharing rules (such as the Loan Officer Role). This allows real-time monitoring, under strict security standards, of the overall status of the loan portfolio and the operational volume of payment receipts.

Visual structure and dashboard components: Dashboard Name: Loan Management Overview (built on top of the Loan Portfolio Summary and Loan Repayment Detail Report reports).

- **Component 1 (Donut Chart — Loan Portfolio Summary):** visualizes the distribution and total sum of the principal loaned ($4.8K), segmented by request status (Approved at $3.0K, Paid Off at $1.8K). Rejected loans are excluded, since they were never disbursed and do not represent real capital exposure — providing an accurate financial reading of the portfolio's volume.
- **Component 2 (Horizontal Bar Chart — Loan Repayment Detail Report):** shows the transactional volume (Record Count) of repayment receipts grouped by operational status (Pending, Draft, No Receipt, Submitted, Approved, and Rejected), facilitating quick identification of bottlenecks in receipt validation.

9.1 Architecture consideration and data refresh

Symptom / Technical challenge: Dashboard components in Salesforce do not instantly and automatically recalculate their metrics when source records are modified, requiring control over how current the displayed data is.

Applied solution: Formal incorporation of a sync review and the conscious use of the manual refresh option, ensuring management evaluates decisions based on the exact latest snapshot of the operational pipeline.

Why this solution: Maintains visual integrity aligned with the transactional database without overloading the platform's background processing resources.

Result / Verification: Dashboard published and verified with the organization's real data (Angela James's and Lewis Webb's active portfolio and receipt pipeline).

10. Cross-Cutting Decisions and Key Learnings

Security and architecture

- **3-layer model (Object Permissions / FLS / Sharing-OWD):** System Context bypasses layers 1 and 2 in both variants; only "Without Sharing" also bypasses layer 3.
- **Subflow pattern for FLS bypass:** see the detail and the four applications of the pattern in Section 3.7.
- **Two subflows, not one, when a screen is involved:** a subflow cannot pause its execution to wait for user interaction.

Flow Builder and data

- Picklist values must formally exist before a Flow can write them.

- Picklists always return text in formulas; they require VALUE() before arithmetic operations.
- Empty collections (such as File Upload's contentDocIds) are not Null — use "Is Empty," not "Is Null."

Reporting

- Combining parent and child objects in the same detail report inflates subtotals if row-level granularity isn't controlled.
- The existence/non-existence of a related record is resolved with a Cross Filter (EXISTS/NOT EXISTS), not a Field Filter on a blank field.
- Dashboard components don't recalculate on their own when the source report changes — they require a manual Refresh.

Interface and configuration

- Field Label vs. API Name: the Label can be changed at any time with no risk; the API Name is fixed from the field's creation.
- Dynamic Forms vs. classic Page Layout are mutually exclusive per section — adding a field to one without the other leaves it invisible on screen, even if it's properly configured in FLS.

11. Project Scope Criteria

The principle of "build only what the client explicitly requested" was applied consistently: whenever a scope question arose (interest earned, reporting cadence, amortization system, principal/interest breakdown per installment), the original case statement was revisited before deciding whether to build it or document it as a future improvement. This avoided both leaving real requirements uncovered and investing effort in unrequested functionality.

12. Known Limitations and Future Improvements

12.1 — Architecture and user experience improvements

- Move "Request a Loan" from the Account Quick Action to the Loan Application tab, using Get Records via $User.Id.
- Centralize the interest rate (0.0046) in a Custom Setting or Custom Metadata Type.
- Enable "Allow submitters to recall approval requests" in the Loan Application Approval Process.
- Stacked chart (Principal vs. Interest) per loan in the Dashboard.
- "Active loans by status" report at the Loan Application level, agnostic of whether it has associated receipts.
- AI-assisted automation for Onboarding and Loan Application intake and approval: an AI agent could manage data capture and duplicate validation in Onboarding, as well as Loan Application intake and approval (the 40%-of-salary rule, active-loan check, "Paid Off" check) — all of these are deterministic business rules. Agentifying Loan Repayment approval is explicitly ruled out, since it requires a human visual comparison between the loaded amount/date and the attached bank receipt (an unstructured document) — the only real control point against fraud or data-entry error in the entire system.

12.2 — Functionality outside the original project's scope

- Support for early/advance installment payments.
- Interest_Remaining__c / Principal_Remaining__c fields for a decomposed outstanding balance.

12.3 — Evaluated and discarded

- Amortization system with capital/interest breakdown per installment (French, German, or other) — discarded as not adding value: the loan's total interest is invariant to the chosen amortization method.

13. Conclusion

The project automates the complete lifecycle of a loan — from employee onboarding, through request and approval, to repayment and automatic closure — covering all of the Management Requirements, the Challenges of the previous system, and the case's Additional Requirements. Every design decision was documented with its business justification, and every scope deviation (functionality not built) was explicitly logged as a future improvement, rather than left ambiguous. The system was fully validated with real users (Angela James and Lewis Webb), with no "Login As" simulations, ensuring that the verified behavior is exactly what an end user of Pure Truth Water Factory would actually experience.

Appendix A — Data Dictionary

Field-by-field detail of the system's custom objects, verified against Object Manager. Fields marked as "unused" still exist on the object (inherited from the managed package) but were removed from the Page Layout and do not participate in any active formula.

A.1 — Loan Application (Loan_Application__c)

Field Label	Field Name (API)	Type	Purpose
Loan Application Number	Name	Auto Number	Loan identifier (e.g., LAN-00003).
Applicant Name	Applicant_Name__c	Master-Detail (Account)	Link to the requesting employee.
Employee Number	Employee_Number__c	Formula (Text)	Cross-object formula: Applicant_Name__r.Employment_Number__c. Automatically travels to the related Account (via the Applicant_Name__c Master-Detail relationship) and projects the employee's ID number onto the request, with no need for anyone to retype it. Ensures data consistency and prevents transcription errors — it's a data-convenience/consistency field, not a security field (different from Employee_User__c, see Section 3.2).
Loan Reason	Loan_Reason__c	Picklist	Reason for the loan.
Loan Amount	Loan_Amount__c	Currency(16,2)	Requested principal.
Loan Start Date	Loan_Start_Date__c	Date	Loan start date.
Loan End Date	Loan_End_Data__c	Date	Calculated end date (name has a factory typo, kept as-is).
Loan Term Months	Loan_Term_Months (picklist choice)	Picklist	Loan term in months; requires VALUE() in formulas.
Monthly Loan Repayment	Monthly_Loan_Repayment__c	Formula (Currency)	Calculated monthly installment.
Total Loan Repayment	Total_Loan_Repayment__c	Formula (Currency)	Principal + total interest to be repaid: (Loan_Amount × 0.0046 × Duration) + Loan_Amount.
Amount Paid	Amount_Paid__c	Roll-Up Summary (SUM)	Sum of Amount__c from Loan Repayment Receipt, filtered by Status = Approved.
Loan Balance	Loan_Balance__c	Formula (Number)	Total_Loan_Repayment__c − Amount_Paid__c.
Total Interest Amount	Interest_Earned__c	Formula (Currency)	Total_Loan_Repayment__c − Loan_Amount__c. Label renamed from "Interest Earned" for clarity; original API Name kept (immutable after creation).
Status	Status__c	Picklist	Submitted / Approved / Rejected / Paid Off.

Field Label	Field Name (API)	Type	Purpose
Loan Duration	Loan_Duration__c	Formula (Number)	Managed package field. In use (in the Total_Loan_Repayment__c formula); hidden from the Layout to avoid duplicating Loan_Term_Months.
Loan Interest Rates	Loan_Interest_Rates__c	Formula (Percent)	Managed package field, fixed at 15%. Unused — removed from the Page Layout.

A.2 — Loan Repayment Receipt (Loan_Repayment_Receipt__c)

Field Label	Field Name (API)	Type	Purpose
Loan Repayment Receipt Name	Name	Auto Number	Receipt identifier (e.g., LRR-000008).
Loan Application Number	Loan_Application_Number__c	Master-Detail (Loan Application)	Loan the receipt belongs to.
Payment Date	Payment_Date__c	Date	Date of the payment.
Receipt Code	Receipt_Code__c	Text(100)	Manual bank reference for the receipt.
Amount	Amount__c	Currency(16,2)	Payment amount.
Payment Type	Payment_Type__c	Picklist	Full Balance / Monthly Installment / Partial Payment.
Status	Status__c	Picklist	Pending (default) / Draft / Submitted / No Receipt / Approved / Rejected.

A.3 — Salary Scale (Salary_Scale__c)

Field Label	Field Name (API)	Type	Purpose
Designation Name	Name	Text(80)	Job title/designation name.
Basic	Basic__c	Currency(16,2)	Base salary.
Housing Allowance	Housing_Allowance__c	Currency(16,2)	Housing allowance.
Transport Allowance	Transport_Allowance__c	Currency(16,2)	Transport allowance.
Other Allowances	Other_Allowances__c	Currency(16,2)	Other allowances.
Gross Payment	Gross_payment__c	Formula (Currency)	Sum of Basic + the 3 allowances; used to validate the 40% loan limit.

Appendix B — Automation Inventory

B.1 — Screen Flows

Name	Version	Description
Employee Onboarding	V1, Active	Guides the Finance Officer through onboarding a new employee: captures data, validates that the Employment Number is not duplicated, creates the Account.
Loan Application Request	V5, Active	Allows any employee to request a loan remotely. Validates that the installment does not exceed 40% of salary and that no active loan exists, before creating the record for approval.
Submit a Repayment	V14, Active	Allows the employee to record a repayment on their approved loan, with a mandatory bank receipt, automatically submitting it for review.

B.2 — Autolaunched Subflows

The four subflows in this table run in System Context With Sharing; the FLS bypass pattern that gives rise to them is documented in Section 3.7.

Name	Version	Description
Loan Application - Create Secure	V2, Active	Creates the Loan Application record, resolving the FLS-blocked write to Loan_End_Data__c.
Create Loan Repayment Receipt (Draft)	V1, Active	Creates the receipt with Status = Draft, resolving the same FLS restriction on Status__c.
Update Loan Repayment Status to Submitted	V1, Active	Updates the receipt's Status to Submitted, once the attached receipt is confirmed, in System Context With Sharing.
Confirm Repayment No Receipt	V1, Active	Updates the receipt's Status to No Receipt, once the attached receipt is confirmed, in System Context With Sharing.

B.3 — Record-Triggered Flows

Name	Version / Type	Description
Loan Application - Auto Close on Full Repayment	V1, Active — Before Save (Fast Field Update)	Updates the Status to Paid Off when Loan_Balance__c reaches 0 or less, enabling the employee for a new loan.
Loan Repayment Receipt - Block Manual Creation	V3, Active — Before Save (Fast Field Update)	Blocks unauthorized manual creation of receipts with a Custom Error outside the guided process (condition: Status Equals Pending).

B.4 — Approval Processes

Process	Entry Criteria	Record Editability	Notes
Loan Application Approval	Status Equals Submitted	Administrators OR the currently assigned approver	Adjusted to allow Lewis Webb to view/act on fields added to the approval screen.
Loan Repayment Receipt: Submit Loan Repayment for Approval	Status Equals Submitted	Administrator ONLY	Intentional design: Lewis doesn't need to edit additional fields, only approve/reject against the attached receipt (Section 6.4).